

VESLYANA, Russian Federation

WMO: 237010

Lat: 62.9985N

Lon: 50.8983E

Elev: 103

StdP: 100.09

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.9	-33.8	-40.0	0.1	-36.7	-36.9	0.1	-33.8	6.5	-9.4	5.6	-8.8	0.7	0	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	28.0	18.7	25.9	17.5	23.9	16.6	20.1	25.5	18.9	23.9	17.8	22.1	2.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	22.3	17.0	12.3	20.8	15.9	11.4	20.3	57.9	25.5	54.1	23.9	50.5	22.3	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.5	5.1	-40.1	30.3	3.4	1.5	-42.6	31.4	-44.6	32.3	-46.5	33.1	-48.9	34.2		
			-39.9	21.4	4.1	1.3	-42.8	22.4	-45.2	23.1	-47.5	23.9	-50.4	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-16.4	-14.1	-7.5	-0.2	7.2	13.0	16.0	12.6	7.7	0.9	-6.7	-11.9
	DBStd	12.53	9.44	8.86	6.50	4.87	5.16	4.53	3.91	4.16	3.58	4.04	7.28	9.50
	HDD10.0	4049	820	674	541	306	121	20	3	21	85	281	499	678
	HDD18.3	6681	1078	907	799	556	347	171	94	184	319	539	749	937
	CDD10.0	449	0	0	0	1	33	109	188	101	17	1	0	0
	CDD18.3	37	0	0	0	0	1	9	21	6	0	0	0	0
	CDH23.3	505	0	0	0	0	38	153	259	55	0	0	0	0
	CDH26.7	112	0	0	0	0	7	42	53	11	0	0	0	0
Wind	WSAvg	2.4	2.1	2.3	2.5	2.6	2.6	2.5	2.2	2.2	2.3	2.5	2.3	2.2
Precipitation	PrecAvg	591	39	29	33	34	47	65	58	78	56	59	48	41
	PrecMax	734	58	59	63	76	89	133	101	158	101	123	82	68
	PrecMin	490	24	9	4	2	17	20	18	36	21	11	10	19
	PrecStd	64	9	14	14	18	19	30	22	28	21	27	16	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	1.9	8.1	16.7	27.9	30.9	30.5	28.6	20.9	11.7	6.5	2.9
		MCWB	-0.2	1.0	3.0	9.1	15.6	19.7	19.6	20.1	15.4	10.7	5.8	2.4
	2%	DB	-0.8	0.5	4.4	11.7	23.7	27.6	28.2	24.7	18.0	9.6	3.1	0.9
		MCWB	-1.1	-0.2	1.0	6.0	14.4	18.6	18.8	18.1	13.3	8.4	2.7	0.3
	5%	DB	-2.7	-0.8	2.3	9.2	20.7	24.8	26.4	22.5	15.7	7.8	1.7	0.1
		MCWB	-3.2	-1.4	0.3	4.3	12.4	16.3	18.0	16.8	12.3	6.9	1.3	-0.3
	10%	DB	-4.6	-2.6	1.0	7.0	17.9	22.3	24.6	20.2	13.8	6.5	0.7	-0.9
		MCWB	-5.1	-3.2	-0.2	3.3	11.2	15.5	17.4	15.8	11.6	5.7	0.3	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	1.1	3.6	10.1	17.6	21.1	22.3	21.1	15.7	11.0	6.1	2.2
		MCDB	0.4	1.6	7.2	14.7	23.4	28.2	26.9	26.0	19.6	11.5	6.7	2.7
	2%	WB	-1.2	0.0	1.6	6.6	15.3	19.5	20.4	19.0	14.1	8.6	2.7	0.5
		MCDB	-0.9	0.6	3.5	10.6	22.3	25.5	25.6	23.7	16.8	9.3	2.9	0.8
	5%	WB	-3.1	-1.3	0.7	5.0	13.3	17.9	19.3	17.7	12.8	7.0	1.4	-0.2
		MCDB	-2.7	-0.9	1.8	8.2	19.8	23.1	24.3	21.1	15.3	7.7	1.7	0.1
	10%	WB	-5.0	-3.1	-0.1	3.8	11.5	16.3	18.3	16.4	11.6	5.7	0.5	-1.3
		MCDB	-4.6	-2.7	1.0	6.2	16.9	21.0	22.9	19.5	13.5	6.5	0.7	-0.9
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.5	10.2	9.2	11.1	10.9	11.3	9.5	7.7	4.3	4.8	6.4
		MCDBR	6.1	5.8	9.2	12.5	17.6	16.1	15.2	13.2	12.6	5.9	3.6	4.1
		MCWBR	5.9	5.4	6.9	7.9	9.3	7.7	6.6	6.8	8.0	5.1	3.5	4.1
	5% WB	MCDBR	5.9	5.3	6.8	10.0	15.7	13.5	12.7	11.7	10.9	5.6	3.5	3.9
		MCWBR	5.9	5.1	5.3	6.8	8.9	7.6	6.6	6.7	7.3	5.1	3.5	3.9
Clear-Sky Solar Irradiance	taub	0.242	0.290	0.322	0.353	0.353	0.358	0.405	0.373	0.336	0.309	0.246	0.228	
	taud	1.951	2.046	2.067	2.138	2.417	2.458	2.325	2.393	2.482	2.342	2.038	1.929	
	Ebn at Noon	355	619	754	818	863	865	804	800	766	622	354	174	
	Edh at Noon	35	71	104	120	100	98	110	94	71	54	31	18	
All-Sky Solar Radiation	RadAvg	0.22	0.90	2.26	3.88	4.96	5.20	5.27	3.59	1.94	0.76	0.22	0.09	
	RadStd	0.03	0.17	0.33	0.44	0.41	0.48	0.45	0.34	0.26	0.09	0.05	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page